IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Received 23 April 2024, accepted 9 May 2024, date of publication 17 May 2024, date of current version 24 May 2024.

Digital Object Identifier 10.1109/ACCESS.2024.3402334

RESEARCH ARTICLE

Netravaad: Interactive Eye Based Communication System for People With Speech Issues

RAJESH KANNAN MEGALINGAM, (Senior Member, IEEE), SAKTHIPRASAD KUTTANKULANGARA MANOHARAN, GOKUL RIJU, AND SREEKANTH MAKKAL MOHANDAS

Humanitarian Technology (HuT) Labs, Department of Electronics and Communication Engineering, Amrita Vishwa Vidyapeetham, Amritapuri, India

Corresponding author: Rajesh Kannan Megalingam (rajeshm@am.amrita.edu)

This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by the Institutional Review Committee.

ABSTRACT People who have speech disability due to apahsia caused by stroke, head injury or brain tumor, amyotrophic lateral sclerosis, cerebral palsy, locked-in syndrome or other motor impairments can use augmentative and alternative communication (AAC) for expressing their mind. Communication based on eye movements is very common for these people. Current systems based on eye gestures or eye tracking are very expensive, require re-calibration very often, slower in operation and might need a human interpreter in many cases. To overcome these limitations, we propose Netravaad, a communication system for people with speech disability to use their eyes to create signs and speak through eyes. Through the eye sign language Netravaani, the user can speak English language using eye signs. These eye signs are captured by a low-cost USB camera which are converted to a word, a phrase, a complete sentence, or even multiple sentences using the Sarani algorithm. The text thus created is displayed on the screen and converted to speech to play the audio via speaker making this interactive multi-modal interface for the benefit of the users. Predictive text feature is implemented to reduce the effort of the users in creating signs for all characters in a word. In addition, the need for human interpreters can be eliminated. Thus, the user can communicate to the caregiver, physician or relative through eyes with a feeling that is same as speaking via mouth. The tests conducted with volunteers from different age groups suggest that, on average, the system achieves an accuracy of 91%, 100%, and 93% in detecting alphabets, words, and numbers, respectively, among individuals aged 26 to 35 years. These findings underscore the robustness of the system, indicating its potential to provide significant benefits to the community of individuals with speech disabilities.

INDEX TERMS Eye gaze, eye gesture, eye sign, eye tracking, predictive text, speech disability.

I. INTRODUCTION

Augmentative and Alternative Communication (AAC) is a boon to people with speech or language problems [1]. AAC supports many modes of communication that can be hand gesture based [2], eye gesture based [3], eye blinks [4], tongue [5], head [6], Brain Control Interface (BCI) [7] etc. But all these modes of communication are not useful for all users. Particularly users who have problems due to apahsia caused by stroke, head injury or brain tumour, amyotrophic lateral sclerosis (ALS), cerebral palsy, locked-in syndrome or other motor impairments cannot use tongue or head or hand etc. for communication with their caregivers. For users with ALS and other motor impairments, eye gestures, eye gaze, eye blinks etc. can be used for communication. Translating eye signs into communicable speech is a real challenge. There are three types of AAC including low technology based, high technology based and non-technical. Writing, drawing,

The associate editor coordinating the review of this manuscript and approving it for publication was Ines Domingues.

© 2024 The Authors. This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License.
For more information, see https://creativecommons.org/licenses/by-nc-nd/4.0/

spelling words by pointing to the alphabet, hand gestures, pointing to images, drawings, words etc. are some of the low technologies based or non-technical AAC. High technology based AAC includes using apps on smartphone or any other electronic gadget like tablets to communicate and using voice enabled computer to recognize gestures etc.

Globally, approximately one billion individuals are afflicted by neurological diseases [8], including stroke, spinal cord injury, and Amyotrophic Lateral Sclerosis (ALS). In advanced stages of these conditions, many patients experience a loss of motor control, rendering them unable to speak or use their hands. Communicating with such individuals becomes a daunting challenge, greatly impacting their quality of life. Due to their limited means of expression, they often endure social isolation, emotional distress, and difficulties accessing information, leading to misunderstandings and misinterpretations. Consequently, there is a pressing need for a device to address this issue. A survey conducted by the All-India Institute of Speech and Hearing, Mysore [9], revealed that speech-related disorders were prevalent in 0.79% of the 28,954 individuals surveyed. Additionally, another study [10] reported a cumulative stroke incidence ranging from 105 to 152 per 100,000 persons per year. These statistics underscore a substantial demand for communication devices tailored to individuals with speech impairments.

In cases of full paralysis, patients often retain control over only their eye movements. Consequently, communication options are typically limited to eye gaze-based methods and Brain-Computer Interfaces (BCI). However, BCIs are not widely adopted due to their complexity, making them challenging for everyday use. External noises can interfere with brain signals, complicating the decoding of multiple commands or words from the patient. Moreover, practical concerns such as privacy, confidentiality, and the risk of infections further hinder the widespread adoption of BCI systems. Therefore, there is a pressing demand for systems based on eye gestures to address these challenges. Certain devices intended for individuals with communication difficulties may not be exclusively designed for communication purposes. Instead, they may serve as replacements for standard keyboard and mouse inputs. However, merely replacing computer input devices may not suffice to facilitate easy communication for patients. Therefore, the development of a separate device equipped with a custom Graphical User Interface (GUI), specifically tailored to communication needs, would greatly benefit users. This specialized device would focus solely on communication features, enhancing usability and accessibility for individuals with speech impairments.

The existing systems and devices for AAC for people with ALS and other motor impairments have several limitations including the speed, cost, interpreters, mobility etc. Tobi Dyanvox [11] uses determine where the user is looking on the screen based on their eye movements and translates to symbols and letters. Eye Transfer Board [12] provides transparent board printed with printed with symbols and an interpreter needs to translate the user's eye gestures on the board into meaningful commands. Tarek et al. [13] introduced an eyeblink-based wearable device, while Kowalczyk and Sawicki [14] proposed a device utilizing blinking and winking-based eye gestures, incorporating an IR LED camera and PC. Vaitukaitis et al. [15] developed gesture recognition through a mobile app, and Zhang et al. [16] presented a method for eye gesture-based selection of alphabets from a smartphone GUI. Rahnama-ye-Moqaddam and Vahdat-Nejad [17] introduced the Eye Type method, employing a webcam, display, and PC. Hornero et al. [18] proposed a microcontroller-based wireless symbol chart and wireless speaker module. Upon reviewing these existing systems, it becomes apparent that none of them define a unique eye sign pattern for the formation of all alphabets in a particular language.

We propose Netravaad, an interactive communication system for people with speech disability to use their eyes to create signs and speak through eyes which is fast, cost effective and doesn't need interpreters. The user can communicate with eye signs in two modes: quickly communicate with the caregiver or relative via commonly used words or with written words and sentences, character by character. Predictive text feature is implemented to reduce the effort of the users in creating signs for all characters in a word and while forming sentences. The sign language created using eye signs in Netravaad is called Netravaani. Using Sarani algorithm, the eye signs captured by a low-cost USB camera are converted into words and/or sentences. In this research work we propose Netravaad and Netravaani, an interactive communication system for users with speech issues and speaking natural language using eyes. The main contributions of this research work to the field of AAC system are:

a) Design and development of Netravaani, collection of unique eye signs for Natural Language alphabets and words (English).
b) Design and development of Sarani, an algorithm to detect the alphabets, words and numbers using eye signs.
c) Design and development of the device for eye sign detection for users with ALS and other motor impairments.

II. RELATED WORK

Patients experiencing speech impairments undergo a loss of their innate speaking capabilities [19]. There are multiple methods in which the patients can express themselves by head, hand and eye gestures, brain signals, Electro Oculography (EOG) [20], [21] etc. These methods are used for control of a device or for communicable speech. Accurate tracking of eyes plays a significant role in the case of eye gesture method. One of the commonly used devices or systems found in the literature for eye tracking is the head-mounted system which employs various methods to track eyes. Challenges in 3D gaze estimation for head-mounted eye tracking systems, focusing on marker-based calibration issues and depth estimation errors are addressed in [22]. The proposed method

employs an RGBD camera for environment structure, utilizing saliency detection for 3D calibration targets. Using a low-cost head-mounted webcam with automatic real-time pupil detection, the study proposed in [23] concludes that the model-based algorithm performs better when the head is moving, achieving gaze estimation within a two-degree benchmark. The article [24] introduces a refraction-aware gaze estimation approach that addresses challenges in accurate gaze estimation using 2-D and 3-D gaze estimation methods, considering corneal refraction effects. The proposed gaze estimation methods demonstrate comparable accuracy to state-of-the-art pupil-center-based methods, mitigating the slippage problem.

The research work in paper [25] presents a flexible and accurate gaze estimation approach that implicitly calibrates gaze patterns, eliminating the need for burdensome personal or device calibration. The study presented in [26] addresses the limitations of existing eye-tracking technology, which primarily focuses on 2D point-of-gaze (PoG) estimation. To enhance accuracy for applications like 3D displays and human-computer interaction, the proposed binocular eye-tracking device integrates stereo stimuli, providing highly precise 3D PoG estimation without additional accessories. Though the head mounted based eye tracking method solves multiple challenges, the users might feel uncomfortable wearing the device for prolonged periods.

Gaze prediction for various applications is becoming common now-a-days. SGaze, a data-driven eye-head coordination model for real-time gaze prediction in immersive Head-Mounted Display (HMD) applications without external hardware is presented in [27]. Another study in [28] introduces DGaze, a CNN-based model for predicting users' gaze positions in dynamic virtual scenes. The research work in [29] proposes an eye-tracking solution for virtual reality (VR) headsets that predict future eye gaze using inertial sensors. Gaze tracking in multi-camera assisted living environments using a neural network regressor based on relative facial key points with accurate and temporally stable gaze predictions is presented in [30]. Challenges remain in some of these methods, particularly those which employ ML/DL algorithms that require dataset and training the algorithms using these datasets. The proposed method overcomes these challenges by employing the image processing algorithm which neither requires dataset nor training.

Eye movements and tracking are used in the several areas including biomedical, automobile, human-computer interaction, virtual reality etc. This study [31] explores the potential of automated eye-tracking technology in neurological disease diagnosis, particularly stroke. This paper [32] introduces a novel approach for recognizing depression using eye-tracking data, focusing on attention patterns without the need for eye movement event identification. Using eye-tracking data to detect high-functioning autism in adults by analysing visual processing differences is proposed in research article [33]. Eye tracking calibration for longitudinal studies in medical exams is evaluated in [34]. The research work [35] evaluates the impact of automated driving on visual risk perception by analysing gaze pattern dispersion during manual and automated driving tasks. The eye tracking based system for AAC proposed in [36] uses a camera to track the eye movement. However, the work is very primitive with evaluation for tracking the eyes but not any useful communication. Another pilot study in [37] proposed cognitive training through the combined use of eye-tracking technology (ETT) and augmentative and alternative communication (AAC) using the Peabody Picture Vocabulary Test (PPVT-IV). The method is suggested for effective communication for people with speech issues but is not validated.

The paper [38] proposes a vision-based system for continuous driver gaze tracking and eyes-off-the-road detection, addressing driver distraction as a key factor in vehicle collisions. The inadequacies in existing gaze-based grasp intention recognition for assistive robots is addressed in the research article [39]. The research work in [40] explores the transformation of a commercial eye-tracker into an alternative peripheral device for human-computer interaction. The research study [41] addresses the challenge of manual map requests for navigation assistance in virtual reality (VR), impacting user experience. There are other applications and approaches for people with speech issues like that of using Electro Oculography (EOG) [42], hand gestures [2], [43], [44], [45], [46] brain computer interface (BCI) [7], [47], [48], [49] for control of devices but not for communicating with others.

Eye gesture-based communication systems are proposed in [3] which uses four eye gestures and encode about 60 daily life commands. A similar method is proposed in [16] but is limited to predicted utterances of predefined words only. There are other eye tracking systems using ML/DL algorithms [50], [51], [52] but are only focussed on development of algorithms for eye tracking and not on eye-based communication for any language. Various eye-based communication languages have been introduced, employing diverse forms of encoding distinct eye gestures. These systems aim to facilitate the seamless and efficient synthesis of communicated speech [13], [53]. Morse-code based approach uses short and long eye blinks corresponding to dots and dashes to form words and sentences [4]. Another eye gesture-based communication system Blink-To-Speak, uses eight eye signs to eight alphabets and create 50 days to day commands for efficient communication [54]. Most of these methods require tailored hardware devices either for the patients or for the caregivers. For example, some of these devices are wearable devices which the patients have to wear like that of Morse-code approach or Eye-Transfer Board [12] which require the caregiver to hold a board. As such they add complexity to the communication process, making it less user-friendly and accessible for both patients and caregivers [55], [56]. In addition, some of these devices are highly priced making them accessible only to high-income countries.

TABLE 1. Different symbols for different eye signs.

Symbol	Eye sign
-	Looking Center
↓	Looking Down / Close
↑	Looking Up
→	Looking Right
←	Looking left

The paper [57] introduces an on-screen keyboard that allows users to communicate exclusively through eye blinking coordination, facilitated by image processing and DLib facial landmarking. The authors utilized a pre-existing model in the DLib toolkit, trained via SVM algorithm. Employing a transfer learning approach enhanced efficiency, yet the achieved accuracy stands at 80% only. Furthermore, the proposed method lacks comprehensive testing and validation. The study [58] concentrates on improving Human-Computer Interaction (HCI) and computer vision frameworks to develop an affordable tracking system tailored for individuals with disabilities. It employs camera input to track the user's pupil in real-time, facilitating tasks like computer mouse pointer control. However, the paper lacks crucial details regarding the test setup, the selection criteria and number of subjects, particularly whether they are individuals with disabilities, the number of tests conducted, and validation procedures.

III. METHODS AND MATERIALS

A. NETRAVAAD AND NETRAVAANI

Netravaad includes a touch display, a camera, a speaker, a mini-PC, a 24V Battery. All these modules are mounted on a portable and adjustable stand, which allows flexibility in setting the camera and display at any height and orientation as per the user's requirement. A unique sign language called Netravaani is defined using five simple, basic eye signs as shown in the Fig. 1 and their combinations. These basic eye signs include center, left, right, up and down. The corresponding symbols are provided in the Table 1. By using various combinations of eye signs the user can create all the English alphabets, words, sentences and numbers. Each combination of eye signs starts and ends with 'center' eye sign so that the user remembers it easily. For example, if the user wants to create the alphabet 'a' then the corresponding eye sign pattern is: center -> up -> right -> center. This can be encoded as [- ↑ → -] pattern as shown in the Table 2. The eye sign patterns for all the 26 alphabets and ten numbers are shown in the Table 2. The eye sign patterns for the numbers are shown in Table 3. The eye signs are captured by the camera and decoded and interpreted into characters, words and/or sentences by using the Sarani algorithm installed in the mini-PC. The speaker is used for the voice output corresponding to the characters, words and sentences. A simple GUI that we developed and installed in the PC gets launched when the system is powered.

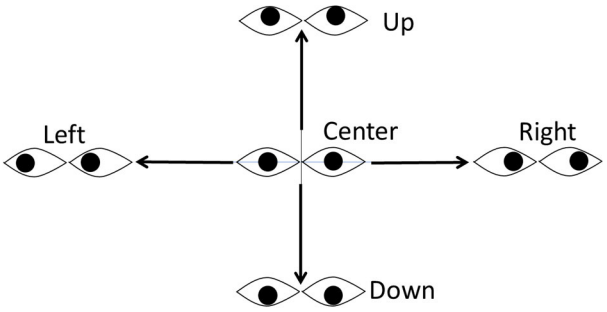


FIGURE 1. Five simple, basic eye signs.

TABLE 2. Alphabets and numbers and their corresponding patterns formed by various combination of basic eye signs.

Alphabet	Pattern	Alphabet	Pattern	Alphabet	Pattern
a	- ↑ → -	k	- ↑ → ↓ -	u	- ↓ ← ↑ → -
b	- → ↓ -	l	- → ↓ ← -	v	- ← ↑ → ↓ -
c	- ↓ ← -	m	- ↓ ← ↑ -	w	- ↑ ← ↓ → -
d	- ←↑ -	n	- ← ↑ → -	x	- ← ↓ →↑ -
e	- ↑ ← -	o	- ↑ ← ↓ -	y	- ↓ → ↑ ← -
f	- ← ↓ -	p	- ← ↓ → -	z	- → ↑ ← ↓ -
g	- ↓ → -	q	- ↓ → ↑ -		
h	- → ↑ -	r	- → ↑ ← -		
i	- ↑ ↓ -	s	- ↑ → ↓ ← -		
j	- ↑ ↓ ← -	t	- → ↓ ← ↑ -		

TABLE 3. Numbers and corresponding pattern.

Number	Pattern
0	- ↑ → -
1	- → ↓ -
2	- ↓ ← -
3	- ←↑ -
4	- ↑ ← -
5	- ← ↓ -
6	- ↓ → -
7	- → ↑ -
8	- ↑ ↓ -
9	- ↑ ↓ ← -

B. NETRAVAAD OPERATION

The user, depending on the situation may be seated in front of the display with a camera mounted over it. If the user is bed ridden, the display with camera can be adjusted in such

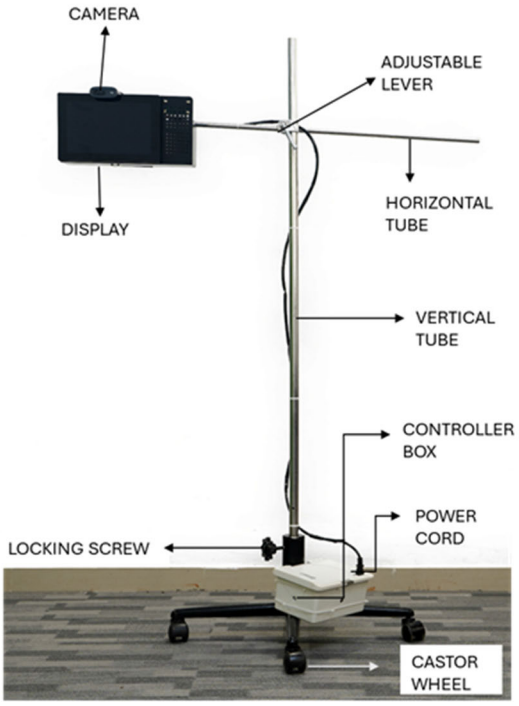


FIGURE 2. The Netravaad system.

a way that the user can face the display. Both the display with camera is mounted on a horizontal rod which can be adjusted either horizontally or vertically with the support of the vertical rod as shown in the Fig. 2.

Beyond this point there is no need for the trainer to be present and the user can independently operate the device using eye signs. The eye signs of the user, shown in Fig. 1, are captured by the camera and processed by the controller for possible detection of alphabets or numbers as given in Table 2 and Table 3 respectively using image processing method. The detected alphabets or numbers are displayed on the screen and also spelled by the speaker attached to the system. Each user should undergo the training phase in which they get familiarized with the eye signs to begin with. The basic eye signs shown in Table 1 are displayed on the display screen for the users for training. The trainers instructs the user to make eye signs as per the signs displayed on the screen. This process is repeated till the users get familiar with the eye signs. Only during training, the presence of the trainer is required. After the training is over, the trainer may stay with the user for a day or two to make sure that the user is able to learn well about using the system. The English 1 mode allows the user to train with predetermined set of words as shown in Table 4. This allows the user to speak the most commonly used words through their eyes using the Netravaad system and also to practice the eye signs.

The Sarani algorithm serves as the core component, orchestrating a sequence of operations to discern user intent through eye movements. It initiates by capturing frames from a webcam and preprocessing them for analysis, which involves cleaning up extraneous elements and resizing for compatibility with the system. Subsequently, it scans for a face within the image, as it holds crucial eye-related cues. Once a face is detected, the algorithm scrutinizes the image to locate the user's eyes with precision. It then employs sophisticated pupil tracking techniques to monitor eye movements accurately. By observing these movements, the system deduces the user's focal points, greatly aiding in comprehension of their intentions. Lastly, leveraging this eye data, the algorithm computes vertical and horizontal ratios, which are utilized to infer gaze directions—left, right, top, center, and close/down.

The various steps involved in Sarani algorithm is listed below:

a. Face Detection: The webcam captures video frames and preprocessing is done using the openCV library. These processed frames to detect faces using a face detection algorithm in Dlib's frontal face detector. Once a face is detected, its position and bounding box are determined.
b. Facial Landmark Detection: After detecting a face, the next step is to detect facial landmarks such as the corners of the eyes, nose, and mouth. This is done using algorithms like DLib's shape predictor using a pretrained model, which predicts the locations of predefined facial landmarks on the detected face.
c. Eye Region Extraction: Once facial landmarks are detected, the algorithm identifies the regions corresponding to the left and right eyes based on the detected landmarks. These regions are isolated from the rest of the face for further processing.
d. Pupil Tracking: Within each isolated eye region, the algorithm locates the position of the pupil. This can be done using techniques such as thresholding to identify the pupil within the eye region. Here the threshold value is set to a particular value throughout the process based on a calibration process which finds the best value.
e. Calibration: In this process the algorithm iteratively evaluates eye frames to determine the optimal binarization threshold for pupil detection. It stores the threshold values for both eyes and provides methods to check the completeness of calibration and retrieve the average threshold values for each eye.
f. Eye Movement Analysis: Once the positions of the pupils are determined, their movements are tracked over time. By analysing the changes in the positions of the pupils relative to the eye frames, the algorithm infers the direction of gaze.
g. Direction Determination: Based on the tracked movements of the pupils and the calibration data, the system determines the direction of gaze. This involves comparing the positions of the pupils relative to the eye centre. The algorithm measures eye movements using the ratio between key point distances within the eye region. It captures both horizontal and vertical gaze.

Horizontal EAR (HR): Analyzes pupil position (x-coordinate) for left/right gaze (0.0-1.0 scale).

Vertical EAR (VR): Analyzes pupil position (y-coordinate) for up/down gaze (0.0-1.0 scale).

Blinking Ratio (BR): Analyses pupil position based on blink to determine the close or down gaze where the gaze is detected when is BR scale should be greater than 4.5.

Sarani defines three different ratios including vertical ratio (VR), horizontal ratio (HR), and blinking ratio (BR) based on the center points of the detected eye and coordinates of detected left and right pupil as in (1) to (10). The five different eye signs are detected based on the values of HR, VR and BR.

$$\mathrm{EC}_{\mathrm{x,y}} = \left(\mathrm{EC}_{\mathrm{x}}, \mathrm{EC}_{\mathrm{y}}\right) = \left(\frac{W}{2}, \frac{H}{2}\right) \tag{1}$$

$$\mathrm{VR}_{\mathrm{Left}} = \frac{P_{ly}}{\mathrm{EC}_{\mathrm{y}} \times 2 - 10} \tag{2}$$

$$\mathrm{VR}_{\mathrm{Right}} = \frac{P_{ry}}{\mathrm{EC}_{\mathrm{y}} \times 2 - 10} \tag{3}$$

$$\mathrm{VR} = \frac{\mathrm{VR}_{\mathrm{Left}} + \mathrm{VR}_{\mathrm{Right}}}{2} \tag{4}$$

$$\mathrm{HR}_{\mathrm{Left}} = \frac{P_{lx}}{\mathrm{EC}_{\mathrm{x}} \times 2 - 10} \tag{5}$$

$$\mathrm{HR}_{\mathrm{Right}} = \frac{P_{rx}}{\mathrm{EC}_{\mathrm{x}} \times 2 - 10} \tag{6}$$

$$\mathrm{HR} = \frac{\mathrm{HR}_{\mathrm{Left}} + \mathrm{HR}_{\mathrm{Right}}}{2} \tag{7}$$

$$\mathrm{BR}_{\mathrm{Left}} = \frac{W_l}{H_l} \tag{8}$$

$$\mathrm{BR}_{\mathrm{Right}} = \frac{W_r}{H_r} \tag{9}$$

$$\mathrm{BR} = \frac{\mathrm{BR}_{\mathrm{Left}} + \mathrm{BR}_{\mathrm{Right}}}{2} \tag{10}$$

The center of the eye $\mathrm{EC}_{\mathrm{x,y}}$ is represented in equation (1) where W is the width of the eye and H is the height of the eye. The x and y coordinates of the eye center is represented as EC_{x} and EC_{y}. The x and y coordinates of the left pupil are represented as P_{lx} and P_{ly}. The x and y coordinates of the right pupil is represented as P_{rx} and P_{ry}. The vertical ratio VR is obtained using vertical ratio of left eye $\mathrm{VR}_{\mathrm{Left}}$ and vertical ratio of right eye $\mathrm{VR}_{\mathrm{Right}}$ as shown in (2), (3), and (4). Similarly, the horizontal ratio HR is obtained using horizontal ratio of left eye $\mathrm{HR}_{\mathrm{Left}}$ and horizontal ratio of right eye $\mathrm{HR}_{\mathrm{Right}}$ as shown in (5), (6), and (7). Blinking ratio BR is obtained by left eye blinking ratio $\mathrm{BR}_{\mathrm{Left}}$ and right eye blinking ratio $\mathrm{BR}_{\mathrm{Right}}$ as shown in (8), (9), and (10).

The images are captured at 20 frames per second, the detection happens at 20 times in each second. To avoid the mis detections, the Sarani algorithm checks for detection of consecutive eye signs multiple times. Two consecutive detections are required for detection of ''LEFT'' and ''RIGHT'' signs. The ''TOP'' needs three consecutive detections whereas for ''CENTER'' and ''DOWN'' needs nine and four consecutive detections. These numbers were found based on the trial-and-error method. Currently a 13-inch LCD screen is used in Netravaad. All the tests are conducted with this screen. The system allows to modify the frames per second and aspect ratio. If the screen size needs to be changed, the system parameters can be modified. Currently this feature is not included in the GUI.

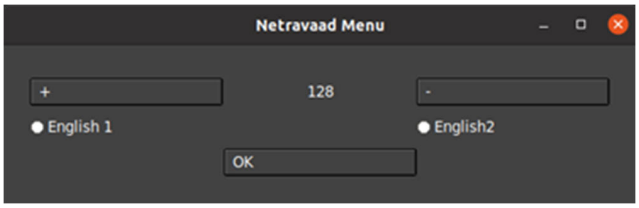


FIGURE 3. GUI for Netravaad.

C. GUI AND CALIBRATION PROCESS

After the powerup, a simple GUI opens up on the touch display of Netravaad. The GUI template is shown in Fig. 3. It has options to choose the English 1 and English 2 modes, and adjust the brightness. English 1 is the default mode in this system. It is to choose a word from a set of predefined word via eye signs. English 2 is for formation of any word or sentence using eye signs. Using the '+' and '-' buttons on GUI the brightness can be adjusted. 128 is the default brightness value. The OK button is used to confirm the selections in the GUI. If the user selects the OK button without adjusting the brightness or selecting a mode, then the default values are taken.

An initial calibration procedure should be completed before the eye sign tracking. When the system is powered up and got connected to Wi Fi network, the GUI guides the calibration process. Calibration includes a feature for adjusting the brightness of the camera input feed and a feature for fixing the head position of the user. The calibration is for the positioning of the face. The device is adjusted in such a way that the user's face is positioned within the red marking as shown in the Fig. 4. During the calibration process, a green rectangle bounding box appears around the user's eye as the eye detection algorithm starts detecting the eyes. The green rectangle bounding box must be within red mark. This step maintains a constant distance between the camera and the user's face and have a straight line of sight with the camera and the user's eyes. To confirm the calibration process, the caregiver can touch on the display. Then a chart of the eye signs corresponding to the selected mode appears on the display.

D. NETRAVAAANI EYE SIGN LANGUAGE

1) MODES OF OPERATION

The user can select between two modes: English 1 and English 2. English 1 is for quick communication with the caregivers, physicians or relatives in which a set of ten predefined, commonly used words can be selected. This mode is also considered as a familiarization mode, useful in getting started with the training of the user before starting with

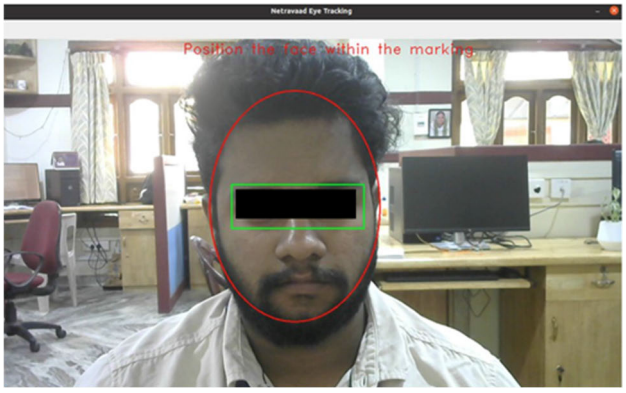


FIGURE 4. User's position of face for calibration of eye sign pattern detection.

TABLE 4. Different patterns for different words.

Pattern	Predefined words
- ↑ -	YES
- ↓ -	NO
- ← -	SIT
- → -	LAY DOWN
- ↑ ↓ -	FOOD
- ↓ ↑ -	SLEEP
- ←→ -	MEDICINE
- →← -	PAIN
- ↑ → -	WASHROOM
- ↑ ← -	WATER

English 2 mode. For leisure communication the user can start the English 2 mode which has four sub-modes: Alphabet mode, Word mode, Sentence mode and Number mode. Each of the sub-modes can be chosen by the user with specific eye signs.

• *English 1 mode:* After selecting English 1 mode, a chart of the eye signs and its corresponding word pops up on the display as in Table 4, so that the user can refer to the chart for eye signs pattern. The user can create the pattern corresponding to the desired word in the list. Once the word is selected, it appears on the screen along with voice for the word. The user confirms the chosen word using the eye sign pattern for 'YES' after which another voice confirmation is issued via speaker and the word selection is completed. For example, if the user chose the word ''SIT'' and confirmed it, then the voice confirmation is, ''YOU HAVE CHOSEN THE WORD SIT''. If 'SIT' is not the word, the user can say 'NO' using eye sign pattern during the voice confirmation and start fresh. Table 4 shows the eye sign patterns and their corresponding predefined words. The pseudo code for the English one mode is also provided in Pseudo Code 1.

• *English 2 mode - Alphabet and word formation:* This mode is to use patterns for alphabets to create words or

Pseudo Code 1 English 1

```
START
WHILE TRUE:
 IF Eye Sign Pattern = Predefined Words THEN
  Display(Predefined Word)
 ELSE IF Eye Sign Pattern = Mode Change THEN
  Display(''Switching to alphabet mode'')
  BREAK
END WHILE
STOP
```

TABLE 5. Patterns for the formation of words.

Pattern	Input
- ↑ -	YES
- ↓ -	NO
- ↑↓ -	LOCK

sentences. When English 2 mode is selected, a chart of the eye sign patterns and its corresponding alphabet pops up on the display as in Table 2, so that the user can refer to the chart for eye signs pattern if needed. Once an alphabet is displayed the user can give two more inputs 'YES' and 'LOCK'. 'YES' can be used to begin the word prediction starting with the chosen alphabet. 'YES' also means the the current alphabet is chosen successful and that the user can switch to the pattern for the next alphabet. The pattern of 'LOCK' can be used to suspend the process for some time. The process can be resumed by giving the same pattern again. 'LOCK' is helpful when the user wants to suspend the Netravaad communication for a brief period and resume later. Table 5 shows the eye sign pattern for YES, NO and LOCK words. The pseudo code for the shared part which is common for word formation, number formation and sentence formation sub-modes using eye sign patterns is shown in Pseudo Code 2.

• *Sentence formation:* Sentence formation mode is selected using the alphabet 'S'. When eye sign pattern for 'S' is performed, the input can be either 'words starting with alphabet S' or the 'Sentence mode'. The pattern 'NO' [-, ↓, -] can be used select the 'Sentence mode'. After selecting the sentence mode, the user can use the same pattern of a–z as in Table 2 to obtain the desired words and create a sentence. Various other eye sign patterns used in sentence formation is shown in Table 6. The user can use the pattern for 'YES' to confirm the alphabet, which is displayed on a separate window. Due to mistake in the pattern if the chosen alphabet is wrong, the pattern 'NO' is used to clear the alphabet. Multiple correct alphabets are concatenated to create words. The pattern for 'SPACE' can be used to add space between words. Instead of creating sentences alphabet by alphabet, the user can choose a sentence from the list of prestored sentences. The Netravaad system is designed in such a way that it gives an option to the user to predict one of the three probable sentences at a time. To select one of the first three sentences from the list, the user

Pseudo Code 2 Shared Pseudo Code

```
START
Menu:
Mode selection
WHILE TRUE:
IF Eye Sign Pattern != ‘‘S’ and ‘N’ THEN
 IF Eye Sign Pattern != Mode Change THEN
// AWP - Alphabet and Word Prediction
 GOTO AWP
 IF Eye Sign Pattern = Mode Change THEN
 Display(‘‘switching to main menu)
 BREAK
 GOTO Menu
 ELSE
 Display(alphabet)
ELSE
GOTO Sentence / Number
END WHILE
STOP
//Pseudo code for Alphabet and Word Prediction (AWP)
AWP:
IF Eye Sign Pattern = YES THEN
  Word suggestion(Alphabet):
  IF Eye Sign Pattern = YES THEN
    Display(Word)
    ELSE
    INCREMENT: word suggestion index
    GOTO Word suggestion
  ELSE
    Alphabet is cleared
```

Pseudo Code 3 Pseudo Code for Sentence

```
  Sentence:
IF Eye Sign Pattern != ‘S’ THEN
GOTO AWP / Number
ELSE
Display(Word Starting with S)
IF Eye Sign Pattern = YES THEN
GOTO Word suggestion (S)
ELSE
Display(Sentence mode)
IF Eye Sign Pattern = YES THEN
  Sentence Mode:
  IF Eye Sign Pattern = Confirm THEN
  Display(Obtained Sentence)
  ELSE IF Eye Sign Pattern = Space THEN
    IF Sentence Prediction available THEN
Select Sentence from prediction and GOTO Sentence Mode
    ELSE
    Append Space and GOTO Sentence Mode
  ELSE IF Eye Sign Pattern = Switch THEN
  Display(Switching to Alphabet mode) and GOTO AWP
  ELSE
    Display(Alphabet)
    IF Eye Sign Pattern = YES THEN
    Alphabet is appended and GOTO Sentence Mode
    IF Eye Sign Pattern = NO THEN
    Alphabet is cleared and GOTO Sentence Mode
ELSE
GOTO AWP
```

can use the patterns [- ← -], [- ↑ -] and [- → -] corresponding to first, second or the third sentence respectively. The user chooses the pattern ‘NO’ to choose from the next three sentences in the list. If no more sentences are available in the list, it changes to manual mode where the user should perform different patterns for each character. To confirm the sentence, the user can input the pattern for ‘CONFIRM’ after which the system provides voice output by reading the sentence the user created. To resume the process, the user needs to give ‘RESUME’ input. After giving ‘CONFIRM’ and it starts a new iteration. To switch to the alphabet formation page user, need to give ‘HOME input’. The pseudo code for sentence formation is shown in Pseudo Code 3.

• *Number formation:* Number formation mode is selected using the alphabet N. The eye sign patterns for numbers are shown in Table 3. When the user creates eye sign pattern for the alphabet N, there are two possibilities. The selection can be either words starting with alphabet N or switching to the number mode. The pattern ‘NO’ [-, ↓, -], can be used to select the number mode. Once the number mode is selected the Table 3 can be used to input the numbers zero to nine. After each number is created, the user can use three different patterns ‘YES’, ‘NO’, and ‘CONFIRM’ as per Table 5 to accept or reject the number. The pattern ‘YES’ [-, ↑, -], indicates that the number is correct and the pattern ‘NO’ [-, ↓, -] indicates that it is a wrong number. In addition, the pattern ‘NO’ clears the number. If the number is correct it is displayed on a separate window. Every time the user choses a correct number, it is concatenated to the previous number. After choosing the required digits, the user can use pattern ‘CONFIRM’ [-, ↓, ↑, -] to confirm the digits as valid. Once the ‘CONFIRM’ pattern is selected the system provides voice output by reading the number (all digits) and starts a new iteration. To switch to the alphabet formation page, the user needs to create the pattern ‘HOME’. The pseudo code for the number formation is shown in Pseudo Code 4.

TABLE 6. Patterns for the formation of sentence.

Pattern	Input
- → ← -	SPACE
- ← -	FIRST
- ↑ -	SECOND
- → -	THIRD
- ↓ ↑ -	CONFIRM
- ↓ -	RESUME
- ← → -	HOME

Pseudo Code 4 Pseudo Code for Number

```
Number:
IF Eye Sign Pattern != 'N' THEN
GOTO AWP/ sentence
ELSE
Display(Word Starting with N)
IF Eye Sign Pattern == YES THEN
GOTO Word suggestion (N)
ELSE
Display(Number mode)
IF Eye Sign Pattern == YES THEN
  Number Mode:
  IF Eye Sign Pattern != Confirm THEN
  Display(Number)
  IF Eye Sign Pattern == YES THEN
  Number is selected and GOTO Number Mode
  ELSE
  Number is cleared and GOTO Number Mode
ELSE
GOTO AWP
```

IV. EVALUATION

A prototype was developed to assess the efficacy of the proposed algorithm Sarani and the system Netravaad. The tests aimed to evaluate accuracy, precision, and recall across various scenarios. These included detection of eye signs at different distances between the device and the user, as well as among different age groups. Participants were informed about the test objectives and provided initial training to familiarize themselves with the device. Two researchers were assigned for conducting the tests: one to give instructions and operate the device, and the other to collect the test results. Following each test, the data was tabulated to calculate the test parameters.

Gaze accuracy may differ among users, and presently, Netravaad utilizes a 13-inch LCD screen for all its tests, which has proven suitable for users. Additionally, the Sarani algorithm includes a provision within the program to adjust frames per second and aspect ratio. However, if there's a need to change the screen size, software developers must manually modify the parameters within the program, as this feature is not accessible via the GUI.

For evaluation of Sarani algorithm, 10 individuals, spanning diverse demographics in terms of age, gender, education, and digital proficiency, were selected as test subjects. They were in the age group between 21 and 25, consisting of 3 females and 7 males. Digital literacy was gauged based on participants' familiarity with utilizing computers, mobile phones, and other digital devices. Each participant underwent a tailored training program lasting two to three days, contingent upon their digital literacy level, prior to the commencement of evaluations. Initial tests were carried out in determining the accuracy of each of the eye signs and their response timings. This is followed by the assessment in completing tasks utilizing English 1 and English 2 modes, encompassing predefined commonly used words, alphabets, and numbers as delineated in Tables 2, 3, and 4. Participants were instructed to employ basic eye signals (left, right, up, down, and center) to articulate the predetermined words, alphabets, and numbers via eye movements. In a similar vein, Netravaad was evaluated through tests conducted on a sample of 10 subjects, comprising 3 females and 7 males in the age group of 21 to 25 years. These tests involved detecting alphabets, words, and numeric at varying distances from the camera. Additionally, assessments were carried out across different age groups: 15 to 25 years (Group G1), 26 to 35 years (Group G2), and 36 to 45 years (Group G3).

TABLE 7. Accuracy in detecting the eye signs.

Sl no	Sub.	Age group	Left	Right	Top	Down	Centre
1	M	G1	0.1	0	1	0.1	1
2	M	G1	1	1	1	1	1
3	F	G1	1	0.9	0.4	0.8	1
4	M	G2	1	0.5	1	1	1
5	M	G2	1	1	1	1	1
6	F	G2	0.9	1	1	1	1
7	M	G3	1	0.8	1	1	1
8	M	G3	0.9	1	1	0.9	1
9	M	G3	1	1	1	0.8	1
10	F	G3	1	1	1	1	1
Average			**0.89**	**0.82**	**0.94**	**0.86**	**1**

TABLE 8. Response time test.

Sl no	Action	Average response time (S)
1	Centre - Top	0.324
2	Top - Centre	0.384
3	Centre - Close	0.292
4	Close - Centre	0.449
5	Centre - Left	0.557
6	Left - Centre	0.164
7	Centre - Right	0.238
8	Right - Centre	0.478

The accuracy in the detection of different eye signs was calculated by asking each subject to perform each of the eye signs 10 times and the average accuracy was calculated as 89%, 82%, 94%, 86% and 100% for the signs Left, Right, Top, Down and Centre respectively as shown in Table 7.

Another test was conducted to determine the response time of detection of different eye signs. The results of the response time test are given below in Table 8. The average time for 10 trails (for each age group) of transition from the "Centre" position was calculated using the video captured during testing. The average value of response time for detection of change from centre to any other eye sign was found to be 0.36 Seconds.

A. EVALUATION OF SARANI

Several tests were carried out to assess the efficacy of the Sarani algorithm in detecting alphabets, words, and numbers. The results of the alphabet detection tests are summarized

TABLE 9. Recall, precision, and accuracy in detecting the correct alphabet.

Sl no	Subject	Age Group	Recall	Precision	Accuracy
1	F	G1	0.83	0.54	0.48
2	M	G1	0.96	0.88	0.85
3	M	G1	0.65	0.58	0.44
4	F	G2	0.88	0.85	0.77
5	M	G2	0.9	1	0.9
6	M	G2	1	0.73	0.73
7	M	G3	1	0.73	0.73
8	M	G3	0.93	0.68	0.65
9	M	G3	0.9	0.63	0.59
10	F	G3	0.86	0.56	0.52

TABLE 10. Recall, precision, and accuracy in detecting the correct words.

Sl no	Subject	Age Group	Recall	Precision	Accuracy
1	F	G1	1	1	1
2	M	G1	1	1	1
3	M	G1	0.96	0.88	0.85
4	F	G2	1	1	1
5	M	G2	1	1	1
6	M	G2	1	1	1
7	M	G3	0.91	0.91	0.83
8	M	G3	1	1	1
9	M	G3	1	1	1
10	F	G3	1	0.89	0.89

in Table 9. Each volunteer underwent 10 trials using the same hardware setup. From these tests, recall, precision, and accuracy metrics were derived to evaluate the system's performance in correctly identifying alphabets. The average values for recall, precision, and accuracy were determined to be 89%, 71%, and 66%, respectively. It was noted that with repeated use of the system, the accuracy of alphabet detection showed potential for improvement.

Additional tests were conducted to assess the performance of the Sarani algorithm in word detection. The objective was to measure recall, precision, and accuracy in correctly identifying words within a 30-second timeframe. The results of these tests are detailed in Table 10. On average, the system achieved recall, precision, and accuracy rates of 98%, 96%, and 95%, respectively. Despite the slightly lower accuracy observed in alphabet detection, the results indicate a higher accuracy in word detection. This enhanced accuracy can be attributed to the word prediction feature integrated into the system. Additionally, users have the capability to clear detected alphabets during the detection process, contributing to improved overall performance in word detection.

The test results for detecting numbers are presented in Table 11. The objective was to assess the system's performance in accurately identifying numbers, measured in terms of recall, precision, and accuracy. On average, the system achieved recall, precision, and accuracy rates of 90%, 93%, and 85%, respectively. Notably, the accuracy of number detection was observed to be higher compared to that of alphabet detection based on these test results.

TABLE 11. Recall, precision, and accuracy in detecting the correct numeric.

Sl no	Subject	Age Group	Recall	Precision	Accuracy
1	F	G1	0.77	0.9	0.71
2	M	G1	0.8	0.89	0.73
3	M	G1	0.93	0.97	0.9
4	F	G2	0.92	0.99	0.91
5	M	G2	0.99	0.98	0.97
6	M	G2	0.98	0.91	0.89
7	M	G3	0.95	0.92	0.87
8	M	G3	0.83	0.94	0.79
9	M	G3	0.99	0.92	0.91
10	F	G3	0.9	0.96	0.86

B. EVALUATION OF THE NETRAVAAD SYSTEM

To evaluate the Netravaad system, another set of 10 volunteers in the age groups between 21 and 25 was chosen. These tests were conducted with the faces of the volunteers positioned at three different distances from the camera: 60cm, 70cm, and 80cm. It was noted that both below 60cm and above 80cm, there was a drastic reduction in recall, precision, and accuracy of the test results. Therefore, distances of 60cm and 80cm were selected as the low and high distance values. For each of these distances, tests were conducted for word, alphabet, and number detection. Ten volunteers, including 3 females and 7 males, participated in these tests. Each volunteer underwent 10 trials using the same hardware setup. The test results for alphabet detection, word detection and number detection at distances of 60cm, 70cm, and 80cm are shown in Fig. 5, Fig. 6 and Fig. 7. These tests aimed to determine the optimal distance between the face and the camera that yields the most accurate output. At the end of the evaluation a questionnaire survey was conducted to get feedback from the users about the system.

From Fig. 5, for alphabet detection we observe that:

a) At a distance of 60cm from the camera, the average recall, precision, and accuracy were 77%, 80%, and 65% respectively.
b) At 70cm away from the camera, the average recall, precision, and accuracy were 89%, 80%, and 73% respectively.
c) At 80cm away from the camera, the average recall, precision, and accuracy were 75%, 71%, and 58% respectively.

Based on these results, it was concluded that 70cm is the optimal distance for alphabet detection with the Netravaad system.

From the Fig. 6, for word detection we observe that,

a) At a distance of 60cm from the camera, the recall, precision, and accuracy were 92%, 97%, and 91% respectively.
b) At 70cm away from the camera, the recall, precision, and accuracy were 98%, 96%, and 95% respectively.
c) At 80cm away from the camera, the recall, precision, and accuracy were 87%, 96%, and 85% respectively.

In this case also we observe that 70cm is the optimal distance for word detection.

From the Fig. 7, for number detection, we observe that:

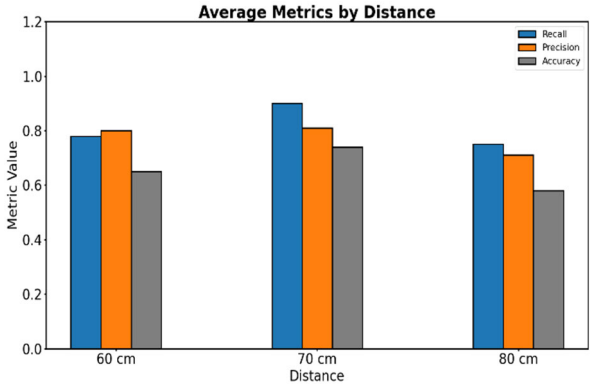


FIGURE 5. Average recall, precision, and accuracy in detecting alphabet, at 60 cm, 70cm, 80cm away from the camera for alphabet detection.

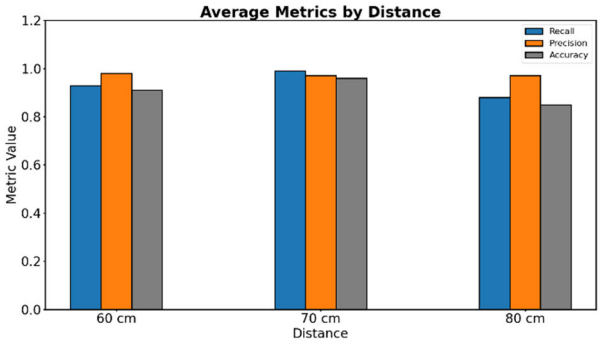


FIGURE 6. Average recall, precision, and accuracy in detecting words, at 60 cm, 70cm, 80cm away from the camera for word detection.

a) At 60cm from the camera, the recall, precision, and accuracy were 87%, 90%, and 79% respectively.
b) At 70cm away from the camera, the recall, precision, and accuracy were 90%, 93%, and 85% respectively.
c) At 80cm away from the camera, the recall, precision, and accuracy were 88%, 92%, and 83% respectively.

In this case also the number detection results are best at 70cm distance.

Final tests were conducted with nine volunteers, distributed across three distinct age groups. Each subject underwent 10 trials using consistent hardware configurations. The age groups were categorized as follows: individuals aged 15 to 25 years comprised the first group, those aged 26 to 35 years constituted the second group, and participants aged 36 to 45 years formed the third group. The tests encompassed detection of alphabets, words, and numbers.

For alphabet detection results given in Table 12 we observe that:

a) The first age group (15 to 25 years) demonstrated a recall, precision, and accuracy of 84%, 78%, and 70% respectively.
b) The second age group (26 to 35 years) exhibited a recall, precision, and accuracy of 92%, 78%, and 91% respectively.

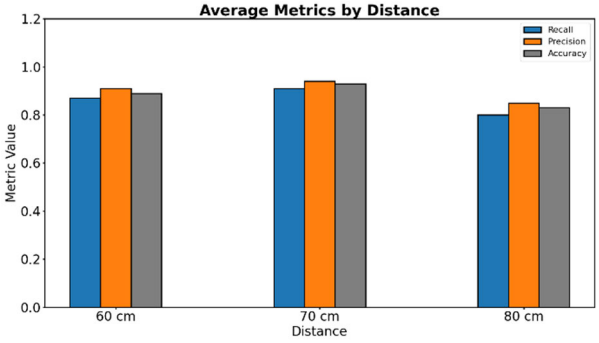


FIGURE 7. Average recall, precision, and accuracy in detecting alphabet, at 60 cm, 70cm, 80cm away from the camera for number detection.

TABLE 12. Recall, precision, and accuracy in detecting the correct alphabet, for the different age groups.

Sl no	Subject	Age group	Recall	Precision	Accuracy
1	M	G1	0.91	0.91	0.83
2	F	G1	0.96	0.88	0.85
3	M	G1	0.65	0.58	0.44
4	M	G2	0.87	1	0.87
5	M	G2	0.98	0.98	0.97
6	F	G2	0.93	0.98	0.92
7	M	G3	0.85	0.88	0.77
8	M	G3	0.84	1	0.85
9	M	G3	0.81	0.91	0.77

TABLE 13. Recall, precision, and accuracy in detecting the correct words, for the different age groups.

Sl no	Subject	Age group	Recall	Precision	Accuracy
1	M	G1	1	1	1
2	F	G1	1	1	1
3	M	G1	0.91	0.91	0.83
4	M	G2	1	1	1
5	M	G2	1	1	1
6	F	G2	1	1	1
7	M	G3	1	1	1
8	M	G3	1	1	1
9	M	G3	0.96	0.88	0.85

c) The third age group (36 to 45 years) displayed a recall, precision, and accuracy of 83%, 93%, and 79% respectively.

Notably, individuals from the second age group (26 to 35 years) demonstrated superior accuracy in alphabet detection compared to the other age groups.

For word detection results given in Table 13 we observe that:

a) The first age group (15 to 25 years) achieved a recall, precision, and accuracy of 96%, 97%, and 94% respectively.
b) The second age group (26 to 35 years) attained perfect scores with a recall, precision, and accuracy of 100% each.
c) The third age group (36 to 45 years) demonstrated a recall, precision, and accuracy of 98%, 95%, and 95% respectively.

TABLE 14. Recall, precision, and accuracy in detecting the correct number, for the different age groups.

Sl no	Subject	Age group	Recall	Precision	Accuracy
1	M	G1	0.8	0.89	0.73
2	F	G1	0.77	0.9	0.71
3	M	G1	0.83	0.94	0.79
4	M	G2	0.99	0.98	0.97
5	M	G2	0.99	0.92	0.91
6	F	G2	0.92	0.99	0.91
7	M	G3	0.93	0.97	0.9
8	M	G3	0.98	0.91	0.89
9	M	G3	0.95	0.92	0.87

For number detection results given in Table 14 we observe that:

a) In the first age group (15 to 25 years), the recall, precision, and accuracy were 80%, 90%, and 74% respectively.
b) In the second age group (26 to 35 years), the recall, precision, and accuracy were 96%, 96%, and 93% respectively.
c) In the third age group (36 to 45 years), the recall, precision, and accuracy were 95%, 93%, and 88% respectively.

Notably, individuals from the second age group (26 to 35 years) demonstrated superior accuracy in number detection compared to the other age groups. From all the tests that are performed we observe that the word detection accuracy is better compared to alphabets and numbers as the system allows predictive text features for the users.

C. COMPARISON WITH OTHER SYSTEMS

In our comparative analysis with six existing methods, as presented in Table 15, it became apparent that no other system currently delineates a distinct eye gaze pattern for forming all alphabets within a language. Our proposed Netravaad system employs a graphical user interface (GUI) displaying alphabet patterns, enabling users to construct an infinite array of words and sentences. Notably, the Netravaad system achieves a commendable 95% accuracy in detecting words within a 30-second timeframe. Upon examination of Table 15, it is evident that our proposed method outperforms methods 1, 3, 4, and 5 in terms of accuracy. However, method 2 exhibits superior accuracy in detection when compared to our system. Nonetheless, it is important to note that method 2 relies on blink and wink patterns of the eye, posing challenges for users due to its complexity. In summary, our Netravaad system presents a novel approach to eye gaze-based communication, offering high accuracy and ease of use compared to several existing methods. While method 2 demonstrates superior accuracy, its reliance on intricate eye movements may hinder user experience, reinforcing the effectiveness and practicality of our proposed solution.

At the end of the final phase of the evaluation, a questionnaire was given to both Group 1 and Group 2 volunteers. Ten volunteers were selected for participation in the questionnaire, divided into two groups: Group 1 and Group 2. Group 1 consisted of experienced participants who had been using Netravaad for over two months, whereas Group 2 comprised individuals who had recently completed training and had been using Netravaad for two to four days. The list of questions is shown below.

1. Is it easy to learn eye signs?
2. Are you comfortable using Netravaad?
3. Is Netravaad easy to use?
4. Should the training time be increased?
5. Are you comfortable with making eye signs?
6. Are you able to make eyes signs for alphabets easily?
7. Are you able to make eyes signs for words easily?
8. Are you able to make eyes signs for numbers easily?
9. Are the guidelines displayed on the screen useful?
10. Is the GUI user friendly?

Table 16 displays the average responses of the 10 volunteers in both groups to the questionnaire. A notable observation is that the percentage of affirmative responses in Group 2 is lower compared to Group 1, likely due to Group 2 participants having only recently started using Netravaad,

TABLE 15. Comparison of Netravaani with other methods.

Sl no	Method	Communication	Accuracy
1	Eyeblink-based wearable device [13]	Modified Morse code chart	87.5%
2	Eyeblink-based device [14]	Blinking and winking-based eye gestures	99.68%
3	Eye gesture recognition with mobile app [15]	Eye gesture-based recognition of 4 eye gaze patterns	60% on a smartphone and 67.3% on laptop.
4	Smartphone with GazeSpeak app [16]	Eye gesture-based selection of alphabets from a GUI	86%
5	Eye Type method [17]	Eye gesture-based selection of alphabets from a GUI	93.8%
6	A microcontroller-based wireless symbol chart [18]	Touch input on a symbol chart	99%
7	Eye sign-based device	Eye gesture based Netravaani language and Sarani algorithm	100% (in detecting words)

TABLE 16. Group 1 and group 2 answers for questionnaire.

Q	Group 1	Group 2
1	Yes – 80%	Yes – 70%
2	Yes – 90%	Yes – 70%
3	Yes – 90%	Yes – 80%
4	Yes – 40%	Yes – 50%
5	Yes – 100%	Yes – 80%
6	Yes – 100%	Yes – 70%
7	Yes – 90%	Yes – 60%
8	Yes – 100%	Yes – 70%
9	Yes – 100%	Yes – 80%
10	Yes – 90%	Yes – 70%

whereas Group 1 members possess over two months of experience with the system. Notably, questions 6, 7, and 9 in Group 1 received 100% affirmative responses, reflecting the expertise gained by these users. Furthermore, 40% of users inquired about increasing training time. Overall, the responses indicate a positive reception of the Netravaad system among volunteers, while also suggesting areas for improvement such as training duration.

V. CONCLUSION

Netravaad is a simple to use device for people with speech issues who can use only their eyes to communicate with others including their relatives and caregivers. It is based on capturing eye signs by a camera which are converted to a word, a phrase, a complete sentence, or even multiple sentences using the Sarani algorithm. The Netravaani eye sign language allows the user to create any word or sentence. The caregivers can hear the eyes of the patients speaking via a speaker which is part of the system. Eye-based speech is also displayed on the screen for the patients to check if they performed the eye signs without any error. The system allows the patient to correct any mistakes in their eye signs. Predictive text feature reduces the effort of the users in creating signs for all characters in a word. In addition, the need for human interpreters can be eliminated. However, some of the limitations include accuracy and delay in eye sign detection. The current approach is also highly light sensitive. The GUI can be improved further to allow easy changes in the system configuration. The future improvements include employing machine learning/ deep learning algorithms by creating a dataset of various eye signs by various users. These algorithms can help in achieving high accuracy results and solve any conflicts in eye signs.

ACKNOWLEDGMENT

The authors would like to thank the Electronics and Communication Engineering Department and the Humanitarian Technology (HuT) Laboratories, Amrita Vishwa Vidyapeetham, Amritapuri Campus for the persistent help and support in conducting this research.

INFORMED CONSENT

This research involves human study, who are researchers at the research lab where this research study was conducted. ''Informed consent was obtained from all individual participants included in the study''. The person depicted agreed to the use of their image.

REFERENCES

[1] M. M. Smith and I. Connolly, ''Roles of aided communication: Perspectives of adults who use AAC,'' *Disab. Rehabil., Assistive Technol.*, vol. 3, no. 5, pp. 260–273, Jan. 2008.

[2] R. K. Megalingam, V. Rangan, P. Veliyara, R. R. Krishna, R. Prabhu, R. Katoch, G. S. A. Koppaka, and R. Sankaran, ''Design, analysis and performance evaluation of a hand gesture platform for navigation,'' *Technol. Health Care*, vol. 27, no. 4, pp. 417–430, Jul. 2019.

[3] M. Ezzat, M. Maged, Y. Gamal, M. Adel, M. Alrahmawy, and S. El-Metwally, ''Blink-to-live eye-based communication system for users with speech impairments,'' *Sci. Rep.*, vol. 13, no. 1, p. 7961, May 2023, doi: 10.1038/s41598-023-34310-9.

[4] K. Mukherjee and D. Chatterjee, ''Augmentative and alternative communication device based on eye-blink detection and conversion to morse-code to aid paralyzed individuals,'' in *Proc. Int. Conf. Commun., Inf. Comput. Technol. (ICCICT)*, Jan. 2015, pp. 1–5.

[5] F. Kong, M. N. Sahadat, and M. Ghovanloo, ''Development and preliminary assessment of an arch-shaped stand-alone intraoral tongue drive system for people with tetraplegia,'' in *Proc. IEEE Biomed. Circuits Syst. Conf. (BioCAS)*, Cleveland, OH, USA, Oct. 2018, pp. 1–4, doi: 10.1109/BIOCAS.2018.8584669.

[6] W. Rahmaniar, A. Ma'Arif, and T.-L. Lin, ''Touchless head-control (THC): Head gesture recognition for cursor and orientation control,'' *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 30, pp. 1817–1828, 2022, doi: 10.1109/TNSRE.2022.3187472.

[7] R. K. Megalingam, B. V. A. Gupta, T. U. Dutt, and A. H. Sushruth, ''Switch and thought controlled robotic arm for paralysis patients and arm amputees,'' in *Proc. IEEE Global Humanitarian Technol. Conf., South Asia Satell. (GHTC-SAS)*, Aug. 2013, pp. 243–248, doi: 10.1109/GHTC-SAS.2013.6629924.

[8] *UN Report on Population Suffering From Neurological Disorders*. Accessed: Mar. 21, 2024. [Online]. Available: https://news.un.org/en/story/2007/02/210312#:~:text=Up%20to%201%20billion%20people, according%20to%20a%20new%20United%20dated%2021/03/24

[9] S. K. Sinha, J. Shivaswamy, A. Barman, D. Seth, D. Seshadri, and S. R. Savithri, ''Prevalence of communication disorders in a rural population at taluq level of Gujarat, India,'' *Clin. Epidemiology Global Health*, vol. 5, no. 2, pp. 73–78, Jun. 2017.

[10] S. Kamalakannan, A. V. Gudlavalleti, V. M. Gudlavalleti, S. Goenka, and H. Kuper, ''Incidence & prevalence of stroke in India: A systematic review,'' *Indian J. Med. Res.*, vol. 146, no. 2, pp. 175–185, 2017, doi: 10.4103/ijmr.ijmr_516_15.

[11] Tobiidynavox. (2023). *Assistive Technology for Communication*. Accessed: Jan. 15, 2024. [Online]. Available: https://us.tobiidynavox.com/

[12] (2023). *Board, E.-T. Eye Transfer Board*. Accessed: Jan. 15, 2024. [Online]. Available: https://www.bridges-canada.com/products/9594-1

[13] N. Tarek, M. A. Mandour, N. El-Madah, R. Ali, S. Yahia, B. Mohamed, D. Mostafa, and S. El-Metwally, ''Morse glasses: An IoT communication system based on Morse code for users with speech impairments,'' *Computing*, vol. 104, no. 4, pp. 789–808, Apr. 2022, doi: 10.1007/s00607-021-00959-1.

[14] P. Kowalczyk and D. Sawicki, ''Blink and wink detection as a control tool in multimodal interaction,'' *Multimedia Tools Appl.*, vol. 78, no. 10, pp. 13749–13765, May 2019, doi: 10.1007/s11042-018-6554-8.

[15] V. Vaitukaitis and A. Bulling, ''Eye gesture recognition on portable devices,'' in *Proc. ACM Conf. Ubiquitous Comput.*, Sep. 2012, pp. 711–714, doi: 10.1145/2370216.2370370.

[16] X. Zhang, H. Kulkarni, and M. R. Morris, ''Smartphone-based gaze gesture communication for people with motor disabilities,'' in *Proc. CHI Conf. Human Factors Comput. Syst.*, May 2017, pp. 2878–2889, doi: 10.1145/3025453.3025790.

[17] R. Rahnama-Ye-Moqaddam and H. Vahdat-Nejad, ''Designing a pervasive eye movement-based system for ALS and paralyzed patients,'' in *Proc. 5th Int. Conf. Comput. Knowl. Eng. (ICCKE)*, Mashhad, Iran, Oct. 2015, pp. 218–221.

[18] G. Hornero, D. Conde, M. Quílez, S. Domingo, M. P. Rodríguez, B. Romero, and O. Casas, ''A wireless augmentative and alternative communication system for people with speech disabilities,'' *IEEE Access*, vol. 3, pp. 1288–1297, 2015.

[19] V. S. Yedavalli, A. Patil, and P. Shah, ''Amyotrophic lateral sclerosis and its Mimics/variants: A comprehensive review,'' *J. Clin. Imag. Sci.*, vol. 8, p. 53, Dec. 2018, doi: 10.4103/jcis.JCIS_40_18.

[20] S. Zarei, K. Carr, L. Reiley, K. Diaz, O. Guerra, P. Altamirano, W. Pagani, D. Lodin, G. Orozco, and A. Chinea, ''A comprehensive review of amyotrophic lateral sclerosis,'' *Surgical Neurol. Int.*, vol. 6, no. 1, p. 171, 2015, doi: 10.4103/2152-7806.169561.

[21] J. Light, D. McNaughton, D. Beukelman, S. K. Fager, M. Fried-Oken, T. Jakobs, and E. Jakobs, ''Challenges and opportunities in augmentative and alternative communication: Research and technology development to enhance communication and participation for individuals with complex communication needs,'' *Augmentative Alternative Commun.*, vol. 35, no. 1, pp. 1–12, Jan. 2019, doi: 10.1080/07434618.2018.1556732.

[22] M. Liu, Y. Li, and H. Liu, "3D gaze estimation for head-mounted eye tracking system with auto-calibration method," *IEEE Access*, vol. 8, pp. 104207–104215, 2020, doi: 10.1109/ACCESS.2020.2999633.

[23] V. Aharonson, V. Y. Coopoo, K. L. Govender, and M. Postema, "Automatic pupil detection and gaze estimation using the vestibulo-ocular reflex in a low-cost eye-tracking setup," *SAIEE Afr. Res. J.*, vol. 111, no. 3, pp. 120–124, Sep. 2020, doi: 10.23919/SAIEE.2020.9142605.

[24] Z. Wan, C. Xiong, W. Chen, H. Zhang, and S. Wu, "Pupil-contour-based gaze estimation with real pupil axes for head-mounted eye tracking," *IEEE Trans. Ind. Informat.*, vol. 18, no. 6, pp. 3640–3650, Jun. 2022, doi: 10.1109/TII.2021.3118022.

[25] H. Zhang, S. Wu, W. Chen, Z. Gao, and Z. Wan, "Self-calibrating gaze estimation with optical axes projection for head-mounted eye tracking," *IEEE Trans. Ind. Informat.*, vol. 20, no. 2, pp. 1397–1407, Feb. 2024, doi: 10.1109/TII.2023.3276322.

[26] J. Sun, Z. Wu, H. Wang, P. Jing, and Y. Liu, "A novel integrated eye-tracking system with stereo stimuli for 3-D gaze estimation," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–15, 2023, doi: 10.1109/TIM.2022.3225009.

[27] Z. Hu, C. Zhang, S. Li, G. Wang, and D. Manocha, "SGaze: A data-driven eye-head coordination model for realtime gaze prediction," *IEEE Trans. Vis. Comput. Graphics*, vol. 25, no. 5, pp. 2002–2010, May 2019, doi: 10.1109/TVCG.2019.2899187.

[28] Z. Hu, S. Li, C. Zhang, K. Yi, G. Wang, and D. Manocha, "DGaze: CNN-based gaze prediction in dynamic scenes," *IEEE Trans. Vis. Comput. Graphics*, vol. 26, no. 5, pp. 1902–1911, May 2020, doi: 10.1109/TVCG.2020.2973473.

[29] A. Satriawan, A. A. Hermawan, Y. F. Luckyarno, and J.-H. Yun, "Predicting future eye gaze using inertial sensors," *IEEE Access*, vol. 11, pp. 67482–67497, 2023, doi: 10.1109/ACCESS.2023.3292411.

[30] P. Her, L. Manderle, P. A. Dias, H. Medeiros, and F. Odone, "Uncertainty-aware gaze tracking for assisted living environments," *IEEE Trans. Image Process.*, vol. 32, pp. 2335–2347, 2023, doi: 10.1109/TIP.2023.3253253.

[31] M. A. Hassan, C. M. Aldridge, Y. Zhuang, X. Yin, T. McMurry, G. K. Rohde, and A. M. Southerland, "Approach to quantify eye movements to augment stroke diagnosis with a non-calibrated eye-tracker," *IEEE Trans. Biomed. Eng.*, vol. 70, no. 6, pp. 1750–1757, Jun. 2023, doi: 10.1109/TBME.2022.3227015.

[32] M. Yang, C. Cai, and B. Hu, "Clustering based on eye tracking data for depression recognition," *IEEE Trans. Cognit. Develop. Syst.*, vol. 15, no. 4, pp. 1754–1764, Dec. 2023, doi: 10.1109/TCDS.2022.3223128.

[33] V. Yaneva, L. A. Ha, S. Eraslan, Y. Yesilada, and R. Mitkov, "Detecting high-functioning autism in adults using eye tracking and machine learning," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 28, no. 6, pp. 1254–1261, Jun. 2020, doi: 10.1109/TNSRE.2020.2991675.

[34] P. Chatelain, H. Sharma, L. Drukker, A. T. Papageorghiou, and J. A. Noble, "Evaluation of gaze tracking calibration for longitudinal biomedical imaging studies," *IEEE Trans. Cybern.*, vol. 50, no. 1, pp. 153–163, Jan. 2020, doi: 10.1109/TCYB.2018.2866274.

[35] C. Liu, E. J. C. Nacpil, W. Hou, Y. Qin, and R. Zheng, "Evaluation of visual risk perception of automated driving tasks by analyzing gaze pattern dispersion," *IEEE Trans. Intell. Vehicles*, vol. 9, no. 1, pp. 775–786, Jan. 2024, doi: 10.1109/TIV.2023.3323340.

[36] S. Adithyan, G. Dhyaneshwar, S. Shivani, V. Karthik, and A. Kumar, "Eye tracking as a form of alternate and augmentative communication for people with speech, language and communication needs," in *Proc. 3rd Int. Conf. Smart Syst. Inventive Technol. (ICSSIT)*, Tirunelveli, India, Aug. 2020, pp. 945–948, doi: 10.1109/ICSSIT48917.2020.9214189.

[37] A. Tost, A. Bachiller, Á. García-Cazorla, I. Medina-Rivera, S. Romero, and M. A. Mañanas, "Electroencephalographic assessment in patients with rett syndrome during cognitive stimulation by means of eye tracking technology and alternative and augmentative communication systems," in *Proc. 45th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Jul. 2023, pp. 1–4, doi: 10.1109/embc40787.2023.10340249.

[38] P. Badgujar and P. Selmokar, "Driver gaze tracking and eyes off the road detection," *Mater. Today, Proc.*, vol. 72, pp. 1863–1868, Jan. 2023, doi: 10.1016/j.matpr.2022.10.046.

[39] B. Yang, X. Chen, X. Xiao, P. Yan, Y. Hasegawa, and J. Huang, "Gaze and environmental context-guided deep neural network and sequential decision fusion for grasp intention recognition," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 31, pp. 3687–3698, 2023, doi: 10.1109/TNSRE.2023.3314503.

[40] E. Cáceres, M. Carrasco, and S. Ríos, "Evaluation of an eye-pointer interaction device for human-computer interaction," *Heliyon*, vol. 4, no. 3, Mar. 2018, Art. no. e00574, doi: 10.1016/j.heliyon.2018.e00574.

[41] S. Zhao and S. Cheng, "Adaptive navigation assistance based on eye movement features in virtual reality," *Virtual Reality Intell. Hardw.*, vol. 5, no. 3, pp. 232–248, Jun. 2023, doi: 10.1016/j.vrih.2022.07.003.

[42] R. K. Megalingam, V. Nandakumar, A. Athira, G. S. Gopika, and A. Krishna, "Orthotic arm control using EOG signals and GUI," in *Proc. Int. Conf. Robot. Autom. Humanitarian Appl. (RAHA)*, Dec. 2016, pp. 1–5.

[43] R. K. Megalingam, V. Rangan, S. Krishnan, and A. B. E. Alinkeezhil, "IR sensor-based gesture control wheelchair for stroke and SCI patients," *IEEE Sensors J.*, vol. 16, no. 17, pp. 6755–6765, Sep. 2016, doi: 10.1109/JSEN.2016.2585582.

[44] M. Rajesh Kannan, M. Deepansh, A. Nitin, and S. Nihil, "Implementation of gesture control in robotic arm using Kinect module," *Appl. Mech. Mater.*, vol. 786, pp. 378–382, Aug. 2015, doi: 10.4028/www.scientific.net/amm.786.378.

[45] R. Megalingam, A. Thulasi, and R. Krishna, "Methods of wheelchair navigation: Novel gesture recognition method," *Int. J. Appl. Eng. Res.*, vol. 7, no. 11, pp. 1687–1690, 2012.

[46] R. K. Megalingam, S. Bandhyopadhyay, G. V. Vivek, and M. J. Rahi, "Hand gesture based wireless robotic arm control for agricultural applications," *IOP Conf. Ser., Mater. Sci. Eng.*, vol. 225, Aug. 2017, Art. no. 012204, doi: 10.1088/1757-899X/225/1/012204.

[47] A. Ahani, M. Moghadamfalahi, and D. Erdogmus, "Language-model assisted and icon-based communication through a brain–computer interface with different presentation paradigms," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 26, no. 9, pp. 1835–1844, Sep. 2018, doi: 10.1109/TNSRE.2018.2859432.

[48] C. Verbaarschot, D. Tump, A. Lutu, M. Borhanazad, J. Thielen, P. van den Broek, J. Farquhar, J. Weikamp, J. Raaphorst, J. T. Groothuis, and P. Desain, "A visual brain-computer interface as communication aid for patients with amyotrophic lateral sclerosis," *Clin. Neurophysiol.*, vol. 132, no. 10, pp. 2404–2415, Oct. 2021, doi: 10.1016/j.clinph.2021.07.012.

[49] J. Katona, T. Ujbanyi, G. Sziladi, and A. Kovari, "Examine the effect of different web-based media on human brain waves," in *Proc. 8th IEEE Int. Conf. Cogn. Infocommunications*, Sep. 2017, pp. 407–412.

[50] Z. Wang, J. Chai, and S. Xia, "Realtime and accurate 3D eye gaze capture with DCNN-based iris and pupil segmentation," *IEEE Trans. Vis. Comput. Graphics*, vol. 27, no. 1, pp. 190–203, Jan. 2021, doi: 10.1109/TVCG.2019.2938165.

[51] W. Li, Q. Dong, H. Jia, S. Zhao, Y. Wang, L. Xie, Q. Pan, F. Duan, and T. Liu, "Training a camera to perform long-distance eye tracking by another eye-tracker," *IEEE Access*, vol. 7, pp. 155313–155324, 2019, doi: 10.1109/ACCESS.2019.2949150.

[52] Y. Sun, J. Zeng, and S. Shan, "Gaze estimation with semi-supervised eye landmark detection as an auxiliary task," *Pattern Recognit.*, vol. 146, Feb. 2024, Art. no. 109980, doi: 10.1016/j.patcog.2023.109980.

[53] S. Pai and A. Bhardwaj, "Eye gesture based communication for people with motor disabilities in developing nations," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jul. 2019, pp. 1–8, doi: 10.1109/IJCNN.2019.8851999.

[54] H. Sane. (2022). *Blink-To-Speak*. Accessed: Jan. 15, 2024. [Online]. Available: https://www.blinktospeak.com/blink-to-speak-guide

[55] H. O. Edughele, Y. Zhang, F. Muhammad-Sukki, Q.-T. Vien, H. Morris-Cafiero, and M. O. Agyeman, "Eye-tracking assistive technologies for individuals with amyotrophic lateral sclerosis," *IEEE Access*, vol. 10, pp. 41952–41972, 2022, doi: 10.1109/ACCESS.2022.3164075.

[56] R. Pugliese, R. Sala, S. Regondi, B. Beltrami, and C. Lunetta, "Emerging technologies for management of patients with amyotrophic lateral sclerosis: From telehealth to assistive robotics and neural interfaces," *J. Neurol.*, vol. 269, no. 6, pp. 2910–2921, Jun. 2022, doi: 10.1007/s00415-022-10971-w.

[57] P. B. Jain, S. Bhat, G. Pujari, V. Hiremath, and C. Deepti, "Eye typing-vision based human activity control," in *Proc. IEEE Int. Conf. Distrib. Comput. Electr. Circuits Electron. (ICDCECE)*, Ballari, India, Apr. 2022, pp. 1–5, doi: 10.1109/ICDCECE53908.2022.9792928.

[58] P. Jaiswal, M. Dhakite, C. Dhule, N. Mungale, S. Wazalwar, and A. R. Deshmukh, "Smart AI based eye gesture control system," in *Proc. 7th Int. Conf. Intell. Comput. Control Syst. (ICICCS)*, Madurai, India, May 2023, pp. 1873–1876, doi: 10.1109/ICICCS56967.2023.10142921.



RAJESH KANNAN MEGALINGAM (Senior Member, IEEE) received the bachelor's degree in engineering from the College of Engineering, Guindy, Chennai, and the master's and Ph.D. degrees from Amrita Vishwa Vidyapeetham University, Kollam, India. He has worked as a VLSI Design and Verification Engineer with various companies, such as STMicro Electronics, and Insilicon Incorporation, Bay Area, California, for six years. He is currently the Director of HuT Laboratories and an Assistant Professor with the Department of ECE, Amrita Vishwa Vidyapeetham, Amritapuri, India. He has seven and half years of work experience and fifteen years of research experience. He is leading several research projects in robotics and automation. He has published more than 140 research papers at various international conferences, journals and book chapters. His research interests include low power VLSI, design for manufacturing, and embedded systems. He has won several awards, including the IEEE Undergraduate Teaching Award 2020, the Outstanding Branch Counselor and Advisor Award from IEEE, NJ, USA, the Outstanding Branch Counselor Award from IEEE Kerala Section, and the Award of Excellence from Amrita University.



SAKTHIPRASAD KUTTANKULANGARA MANOHARAN received the M.Tech. degree in applied electronics and communication systems from the Thejus Engineering College, Thrissur, India, in 2015. He is currently pursuing the Ph.D. degree with the Humanitarian Technology (HuT) Laboratories, Department of Electronics, Amrita University, Amritapuri, India. His Ph.D. thesis focuses on object recognition, identification, and depth estimation on a 3-D plane involving non-deterministic structures. He has three years of research experience and one year of teaching experience. His research interests include robotics, computer vision, machine learning, image processing, and power-efficient solar energy systems.



GOKUL RIJU received the Bachelor of Computer Application degree from Amrita Vishwa Vidyapeetham, Amritapuri, in 2021. He is currently a Softer Developer with HuT Laboratories.



SREEKANTH MAKKAL MOHANDAS received the B.Tech. degree in electronics and communication engineering from the Sahrdaya College of Engineering and Technology, Thrissur, India, in 2014, and the M.Tech. degree in very large-scale integration (VLSI) design from Amrita Vishwa Vidyapeetham, Amritapuri, India, where he is currently pursuing the Ph.D. degree. His research interests include robotics, haptics, and embedded system design.